

Sahithi Maram

Software Engineer | Distributed Systems & Cloud Architecture

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EDUCATION

Master's in Computer Science

08/2021 – 08/2022 | Dayton, OH, USA

University of Dayton, OH (GPA: 3.74/4.00)

Bachelor's in Computer Science

08/2015 – 05/2019 | Telangana, India

Bhoj Reddy Engineering College for Women, Hyderabad (GPA: 3.6/4.00)

Coursework: Object Oriented Design, Advanced Algorithms, System Software, Advanced Programming and Data Structures, Database Management Systems-II, Network Security, Operating Systems-II, Data Science & Pattern Recognition, Advanced Intelligent Systems & Deep Learning, Software Engineering.

PROFESSIONAL EXPERIENCE

Software Development Engineer at Amazon, Seattle, WA

08/2022 – Present

- **Tech Leadership:** Served as Tech Lead, driving multiple high-impact projects including the **Dimensions Platform**, **Inbound Physical Buffer Tracking expansion**, and **Physical-Virtual Mismatch Mitigation**, improving planning accuracy, scalability, and data integrity across fulfillment operations.
- **Dimensions Platform:** Designed and launched a scalable multi-type dimension model for distributed fulfillment systems, processing 1M+ dimension records daily. **Improved flow planning accuracy and boosted throughput by 35%** through optimized data structures and caching strategies, enabling real-time decision-making across 20+ fulfillment centers.
- **Physical-Virtual Mismatch Mitigation:** Architected distributed data lifecycle management system using DynamoDB TTL and AWS Lambda, processing millions of stale records with automated cleanup ensuring storage efficiency and data consistency. Implemented idempotent retry logic and eventual consistency patterns, **eliminating 100K+ invalid entries** and **improving dashboard accuracy by 40%**.
- **Inbound Physical Buffer Tracking:** Architected and deployed distributed buffer tracking system across 15+ fulfillment sites, processing 10M+ inventory events daily with sub-second latency. Built idempotent ingestion pipeline with DynamoDB and Lambda, implementing TTL-based data lifecycle management and eventual consistency patterns. Designed horizontally scalable architecture handling 5K+ TPS, **enabling real-time insights for workforce allocation** and demonstrating extensible **distributed systems design requiring no major redesigns for new site integrations**.
- **CI/CD & Release Engineering:** Co-led the migration to **Full Continuous Deployment** with staged rings, automated canaries, quality gates, and auto-rollback, significantly increasing release velocity and reducing manual toil. Designed and rolled out **dual-track beta/main pipelines**, streamlining peak-season operations and enabling fast, compliant emergency patches - saving the team **~8 weeks of development time annually**.
- **Performance & Scale Optimization:** Led performance tuning of Tier-1 distributed services in fulfillment operating system (fOS). Designed and executed large-scale load tests simulating realistic traffic patterns, identified bottlenecks in data access layers, and implemented caching strategies and query optimizations. Scaled system from 6K to 50K TPS while maintaining p99 latencies. Enhanced observability with structured logging, metrics, and alarms for faster incident detection.
- **Talent Development:** Mentored engineers and interns, improved onboarding processes, and raised team-wide standards for **code quality, testing, and observability**.

Software Engineer at Wipro Technologies, India

09/2019 – 08/2021

- Developed and maintained features for Ericsson's networking tools platform, improving frontend responsiveness and enhancing CI pipeline reliability.
- Partnered with QA to design and implement automated test suites, strengthening release hardening and reducing production regressions.
- Designed and implemented telecom networking protocol handlers in C++ and Java, focusing on performance optimization and memory management for large-scale distributed network systems handling enterprise traffic.
- Collaborated with cross-functional teams in an Agile environment, improving product stability and accelerating delivery cycles.

SKILLS

Languages & Technologies: C++, Java, Python, JavaScript, TypeScript, Go, SQL, NoSQL, HTML, Linux, React, Node.js, REST APIs

Distributed Systems: Large-scale infrastructure, Cloud storage systems, Data consistency, High availability, Microservices architecture, TCP/IP

Cloud & Tools: AWS (DynamoDB, Lambda, CloudWatch, AppConfig, CDK, CloudFormation, S3, EC2, SQS), Amazon Coral, Guice, Git, Docker

Practices: System Design (HLD & LLD), Performance tuning at scale, CI/CD, TDD, Observability, Algorithms & Data Structures, Object-Oriented Design, Design Patterns, Agile/Scrum

AI & Productivity: AWS Foundations of Prompt Engineering (Certified), AI-assisted development (Claude/Amazon CodeWhisperer), Prompt engineering for code generation